

MOHAMAD MOHAMAD

Artificial Intelligence & Machine Learning Engineer

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AI Engineer with 4+ years of professional software engineering experience, specializing in **Generative AI** and **enterprise LLM systems**. Combines strong **system architecture** skills with deep knowledge of **RAG**, **Agentic AI**, and transformer-based models. Experienced in integrating AI solutions into production enterprise environments including **SAP** and **ServiceNow** ecosystems.

KEY SKILLS & TECHNOLOGIES

- **Generative AI & LLMs:** RAG, Agentic AI, LangChain, LangGraph, LangSmith, Gemini LLM, LLM Orchestration, Prompt Engineering, Embeddings, Vector Search, Tool Calling, Structured Output Parsing, Transformer Architectures, Vector Databases (Chroma, FAISS, Pinecone)
- **Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, CNNs, RNNs, LSTMs, Attention Mechanisms, NLP, Model Optimization
- **Programming & Frameworks:** Python, PyTorch, Scikit-learn, NumPy, Pandas, SQL, FastAPI, Microsoft Bot Framework
- **Cloud & AI Platforms:** Microsoft Azure (Azure OpenAI, Azure AI Services), Google Cloud Platform (Vertex AI, Gemini API, Cloud Functions, Cloud Run, Firebase), REST API Integration, CI/CD Pipelines

WORK EXPERIENCE

AI & ML Engineer / TCS Hungary

Jun 2025 - Present

- Developed Retrieval-Augmented Generation (RAG) systems using **LangChain**, **LangGraph**, and Large Language Models (LLMs).
- Implemented **Agentic AI architectures** with tool calling, multi-step reasoning, and workflow orchestration.
- Integrated AI applications with **SAP SuccessFactors** and **ServiceNow REST APIs** for enterprise automation.
- Built prompt-engineered solutions using zero-shot, one-shot, and few-shot prompting techniques.
- Developed automation scripts and backend services in **Python**, utilizing Azure AI services and Git-based CI/CD workflows.
- Worked with **Microsoft Bot Framework** to build conversational AI interfaces.
- Integrated and evaluated **Google Gemini LLM APIs** for prototyping AI-powered features such as content generation and conversational workflows.
- Used **LangSmith** for tracing, debugging, and evaluation of LLM workflows.

Senior Mobile Developer (Previous Role)

Oct 2021 - Jan 2025

- Delivered scalable production applications serving real-world users.
- Designed modular architectures and optimized performance-critical systems.
- Led development workflows using CI/CD, backend integration, and version control best practices.

PROJECTS

AI-Powered Recruitment Intelligence Platform | TCS Hungary

- Architected and developed an AI-assisted recruitment intelligence platform combining **LLMs**, **semantic retrieval**, and **deterministic scoring** to automate candidate ranking, cross-job matching, and interview workflows.
- Designed multi-stage **RAG** and **Agentic AI** pipelines using **LangChain**, vector databases (**Chroma/FAISS**), and Local LLM Deployment (**Ollama, LM Studio**).
- Implemented **semantic capability extraction** and matching pipelines that evaluate candidates beyond keyword similarity using **embeddings**, capability relations, and **explainable scoring methodologies**.
- Built a **hybrid ranking engine** combining LLM-based overlap estimation with **deterministic calibration** formulas, **weighted capability matching**, and **relevance-gated experience scoring**.
- Developed asynchronous **FastAPI** backend services for resume ingestion, job parsing, candidate ranking, **cross-ranking** across multiple job descriptions, and automated interview communication workflows.
- Engineered structured **JSON validation and repair pipelines** to improve reliability and robustness of **LLM-generated outputs** in production workflows.
- Integrated **vector similarity search**, **configurable scoring systems**, and **explainable AI-generated fit summaries** to support transparent hiring decisions and enterprise scalability.

AI-Powered HR Support & Ticket Automation System | TCS Hungary

- Developed an enterprise AI workflow automation system that ingested HR support emails, classified service requests using **LLM-based intent analysis**, and automatically created structured **ServiceNow** tickets with contextual metadata and categorization.
- Designed **AI orchestration pipelines** that retrieved relevant information from enterprise documents, **SAP APIs**, and email attachments to support automated issue resolution workflows.
- Implemented **human-in-the-loop** AI response generation, producing formatted draft resolutions for HR teams to review, edit, and send to clients.
- Built **semantic retrieval** and enterprise integration workflows to improve support efficiency, reduce manual ticket triage, and accelerate issue handling processes.
- Integrated **AI-powered classification**, retrieval, and workflow automation into production enterprise environments using **Python-based backend services** and **REST APIs**.

EDUCATION

- **MSc Computer Science (Artificial Intelligence)**
Eötvös Loránd University (ELTE), Budapest | 2024 - 2027
- **BSc Information Technology Engineering (AI Specialization)**

ADDITIONAL INFORMATION

- **Languages:** English (C1), German (A1 - currently improving), Arabic (Native)
- **Hobbies:** Reading, Running, Hiking, Volleyball.